Causal Inference Homework 2

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1 Abadie et al. Synthetic Control Model

1. Estimate a synthetic control on the California smoking data. Use the same model as in the Abadie et al JASA paper

Looking at the Abadie et al. paper, I see they average the predictor variables over 1980-1988 as well as add three lags of smoking consumption (1975, 1980, 1988). Doing this in my model (and utilizing the nested option in synth) I get the result presented in Figure 1. This figure appears to be very similar to the one presented in the Abadie et al. paper. I noticed in the paper they also move the line representing treatment year from 1989 to 1988 so that the line is placed before the first treated year. I have done the same in the above graph. Interpreting the results, we can see a clear deviation between real California and its counterfactual in the aftermath of the treatment. This suggests the treatment had a significant negative impact on cigarette consumption - around 20 packs per capita annually.

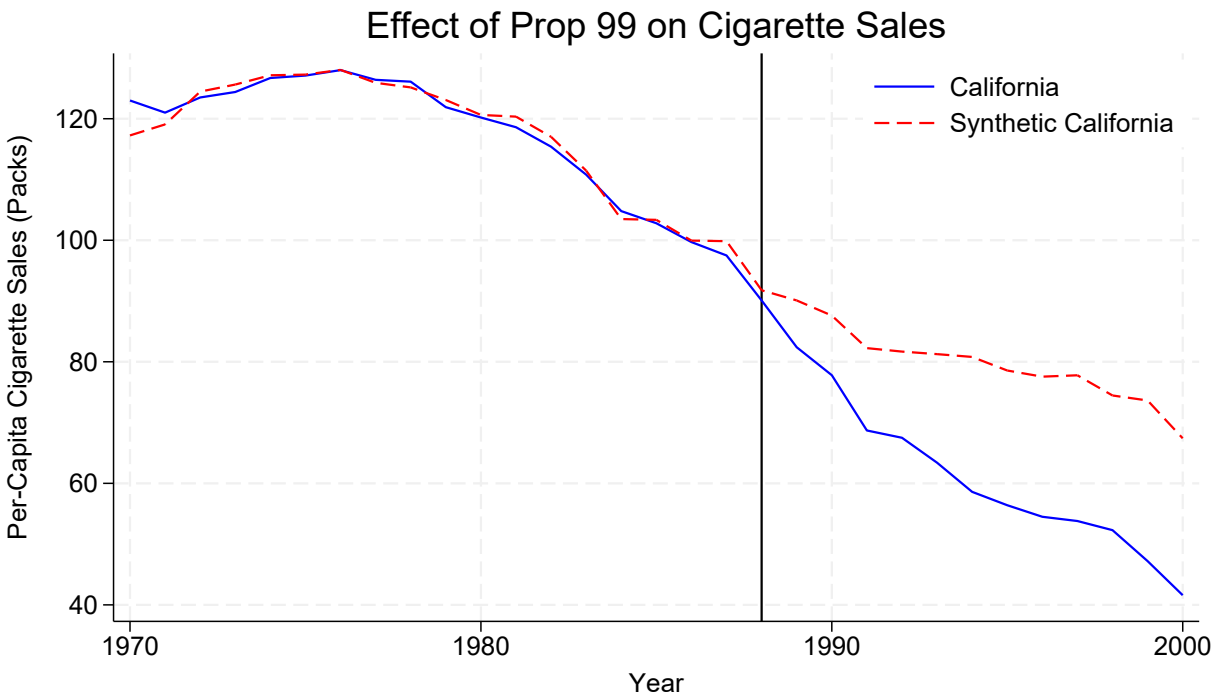

Figure 1: Abadie et al. Replica Synthetic Control

2 Abadie et al. Synthetic Control Model Tables

2. Show your results. What is the composition of the syn. Control? How do the indicator variables match up? What does the treatment effect look like?

The composition of the synthetic control is given in Table 1 below. The weights are almost entirely identical as in the Abadie et al. paper (it should be notes that without specifying the "nested" option in synth it did not give the same weights). Similarly, predictor balance is given in Table 2. The balance is very similar to those presented in the Abadie et al. paper, and, if anything, these results show a slightly better balance.

Table 1: Synthetic Control Unit Weights

State	**Weight**	**State**	**Weight**
Alabama	0	Nevada	0.236
Arkansas	0	New Hampshire	0
Colorado	0.16	New Mexico	0
Connecticut	0.068	North Carolina	0
Delaware	0	North Dakota	0
Georgia	0	Ohio	0
Idaho	0	Oklahoma	0
Illinois	0	Pennsylvania	0
Indiana	0	Rhode Island	0
Iowa	0	South Carolina	0
Kansas	0	South Dakota	0
Kentucky	0	Tennessee	0
Louisiana	0	Texas	0
Maine	0	Utah	0.335
Minnesota	0	Vermont	0
Mississippi	0	Virginia	0
Missouri	0	West Virginia	0
Montana	0.202	Wisconsin	0
Nebraska	0	Wyoming	0

Table 2: Predictor Balance

Predictor Variable	**Treated**	**Synthetic**
lnincome(1980(1)1988)	10.07656	9.868487
beer(1984(1)1988)	24.28	24.25306
age(1980(1)1988)	0.1735324	0.173699
retprice(1980(1)1988)	89.42222	89.51086
cigsale(1975)	127.1	127.2612
cigsale(1980)	120.2	120.6232
cigsale(1988)	90.1	91.77

3 Synthetic with lagged values

3. Now estimate the model using all the possible lagged outcome variables and no other covariates. What is the composition of the syn. Control? What does the treatment effect look like? How, if at all, do these results differ from the ones in question 2.

Setting up the synthetic control with only lagged values of cigarette consumption gives the result in Figure 2. If anything, the only change visible in the graph is that this option gives an even better pre-treatment fit than using the other covariates. It does also change the weights for the synthetic control, which are presented in Table 3. New Hampshire gets some weight but all the other states remain the same. The predictors are presented in Table 4, showing it has a near-identical match between the synthetic and real California in the pre-treatment in the outcome variable. As can be seen from the graph in Figure 2, the treatment still hovers around a 20 pack per capita decrease in Cigarette consumption.

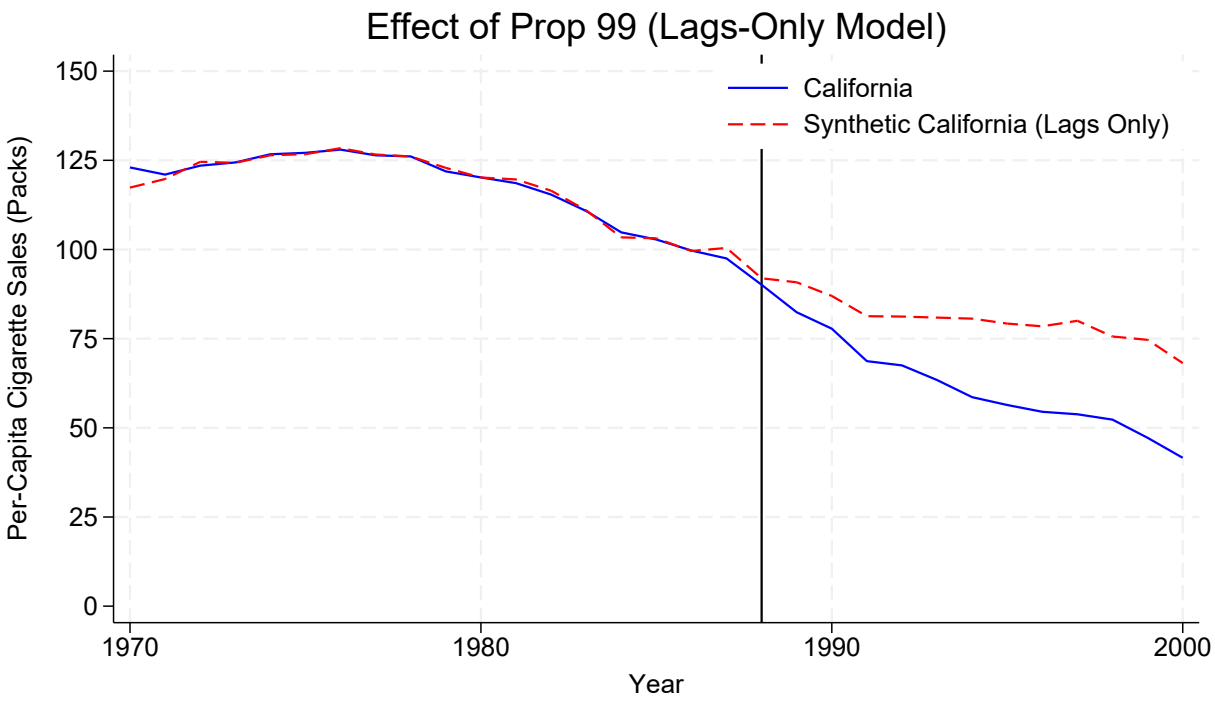

Figure 2: Synthetic California with all lagged outcome variables

Table 3: Synthetic Control Unit Weights

State	Weight	State	Weight
Alabama	0	Nevada	0.205
Arkansas	0	New Hampshire	0.045
Colorado	0.015	New Mexico	0
Connecticut	0.109	North Carolina	0
Delaware	0	North Dakota	0
Georgia	0	Ohio	0
Idaho	0	Oklahoma	0
Illinois	0	Pennsylvania	0
Indiana	0	Rhode Island	0
Iowa	0	South Carolina	0
Kansas	0	South Dakota	0
Kentucky	0	Tennessee	0
Louisiana	0	Texas	0
Maine	0	Utah	0.394
Minnesota	0	Vermont	0
Mississippi	0	Virginia	0
Missouri	0	West Virginia	0
Montana	0.232	Wisconsin	0
Nebraska	0	Wyoming	0

Table 4: Predictor Balance (All Lags Model)

Predictor Variable	Treated	Synthetic
cigsale(1970)	123	117.3614
cigsale(1971)	121	119.7564
cigsale(1972)	123.5	124.5763
cigsale(1973)	124.4	124.3053
cigsale(1974)	126.7	126.4153
cigsale(1975)	127.1	126.7159
cigsale(1976)	128	128.4025
cigsale(1977)	126.4	126.6217
cigsale(1978)	126.1	126.0853
cigsale(1979)	121.9	122.8776
cigsale(1980)	120.2	120.1817
cigsale(1981)	118.6	119.6467
cigsale(1982)	115.4	116.4915
cigsale(1983)	110.8	110.9973
cigsale(1Two-sided4)	104.8	103.4059
cigsale(1985)	102.8	103.1442
cigsale(1986)	99.7	99.5867
cigsale(1987)	97.5	100.4232
cigsale(1988)	90.1	91.9269

4 In-Time Placebo

4. Do an in-time placebo. Assume that prop 9 passed in 1981 and use 82-88 for the treatment period. Show your results. What is the composition of the syn. Control? How do the indicator variables match up? What does the treatment effect look like?

Doing an in-time placebo with a fake treatment at 1981 with the original Abadie et al. model (except beer is dropped because its data starts in 1984 and the lagged outcome value of 1985 is dropped because it is post-treatment) gives the results in Figure 3. As you can see the synthetic and actual California show a slight deviation in the latter half of the 1980's, suggesting there may be a small confounding shock that could be slightly exaggerating our results from previous analyzes. The weights and predictor balances are also presented in Table 5 and Table 6, respectively.

Table 5: Synthetic Control Unit Weights

State	Weight	State	Weight
Alabama	0	Nevada	0.291
Arkansas	0	New Hampshire	0
Colorado	0.047	New Mexico	0
Connecticut	0.318	North Carolina	0
Delaware	0	North Dakota	0
Georgia	0	Ohio	0
Idaho	0	Oklahoma	0
Illinois	0	Pennsylvania	0
Indiana	0	Rhode Island	0
Iowa	0	South Carolina	0
Kansas	0	South Dakota	0
Kentucky	0	Tennessee	0
Louisiana	0	Texas	0
Maine	0	Utah	0.345
Minnesota	0	Vermont	0
Mississippi	0	Virginia	0
Missouri	0	West Virginia	0
Montana	0	Wisconsin	0
Nebraska	0	Wyoming	0

Table 6: Predictor Balance (In-Time Placebo)

Predictor Variable	Treated	Synthetic
lnincome(1972(1)1980)	9.987636	9.884742
age(1970(1)1980)	0.1838485	0.1836236
retprice(1970(1)1980)	47.58182	47.99077
cigsale(1975)	127.1	127.0648
cigsale(1980)	120.2	121.1977

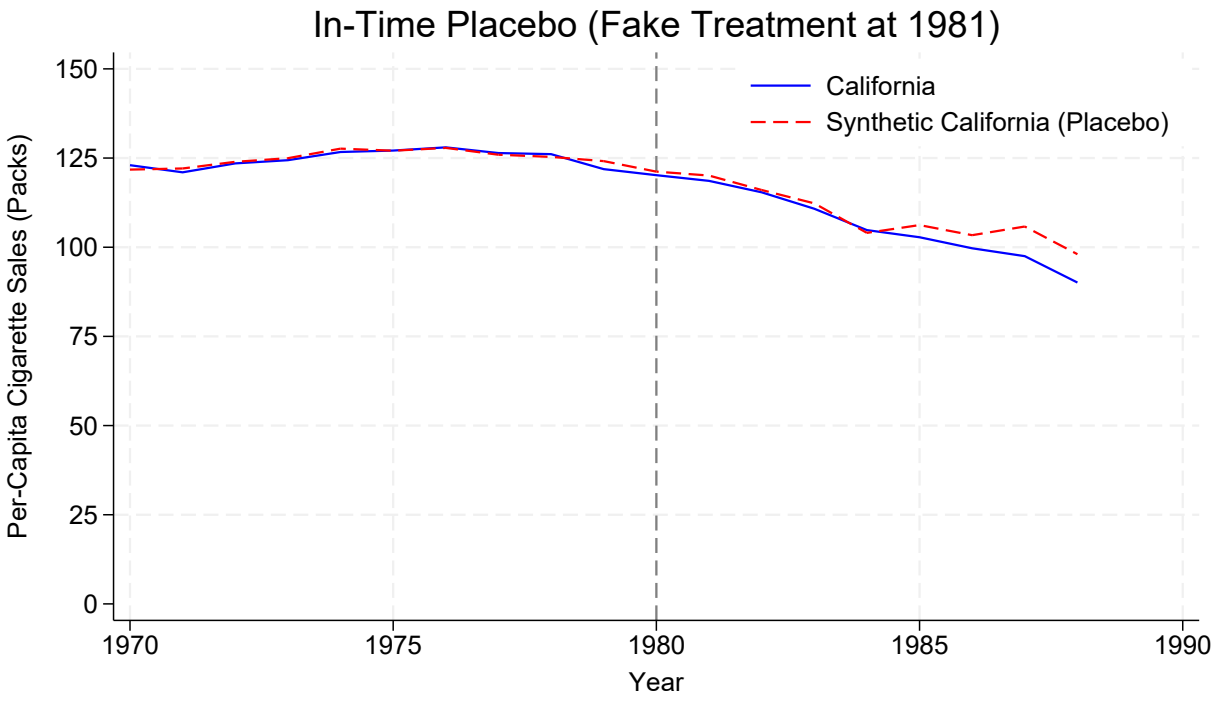

Figure 3: In-Time Placebo: fake treatment in 1981

5 In-Space Placebo Testing

5. Use the synthrunner module (from the resources page) and run placebos to estimate p-values for the models in questions 2,3, and 4.

5.1 Abadie et al. Model

Using synth_runner I perform placebo tests for the 38 control states. The standardized p-values for the significance of California's response to the treatment is given in Figure 4. Standardizing the p-values takes into account the poor pre-treatment fit of the placebos. In this model that recreates the method of Abadie et al. it can be seen that with the standardized p-values are all under the 0.05 threshold. This means we can be 95% confident that the drop we see in cigarette consumption post-treatment is a valid inference. I also plotted the placebos in Figure 5. Following the approach of Abadie et al. in their paper, I drop the placebos that have a MSPE that is greater than two times that of California in the pre-treatment period - this simply allows for easier readability in the plot. This plot shows us that the real California has a much more pronounced affect from the treatment than the placebos (comfirming what we already know).

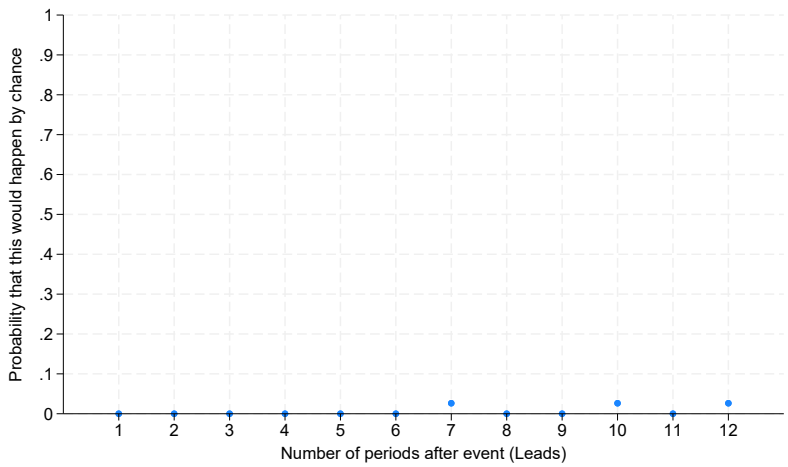

Figure 4: p-value plot

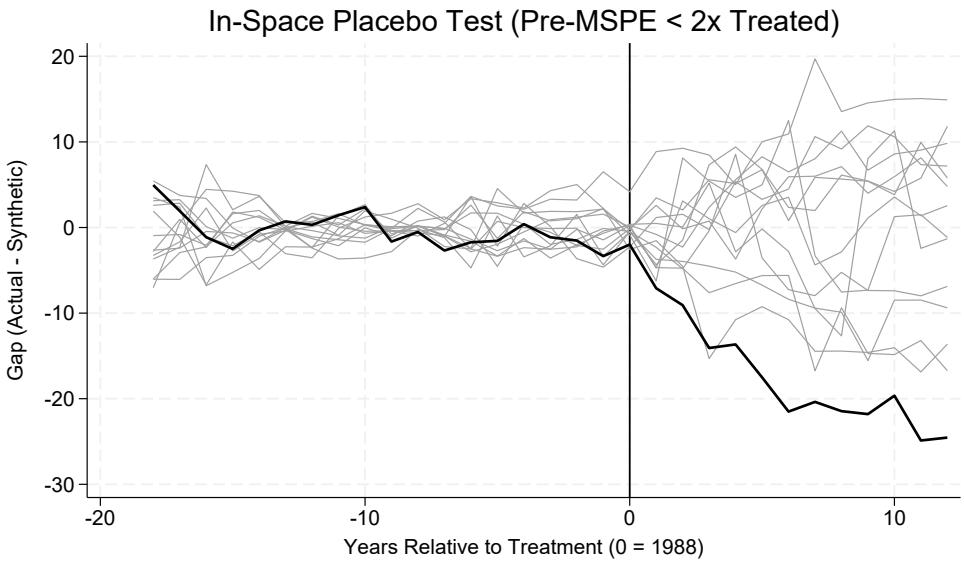

Figure 5: Placebo Plots (less than two times pre-treatment MSPE)

5.2 All lagged outcome variables Model

Using the same method described above but for the all lags model, I calculate the placebos for the 38 control states. Figure 6 shows us that there are actually a couple of years in which the treatment is not significant at the 10% level. However, most states hover around the p=0.05 level, telling us the treatment is generally significant at the 95% confidence level. I plot the placebos again in Figure 7, and this again shows that the real treated California has a much more pronounced effect than the placebos.

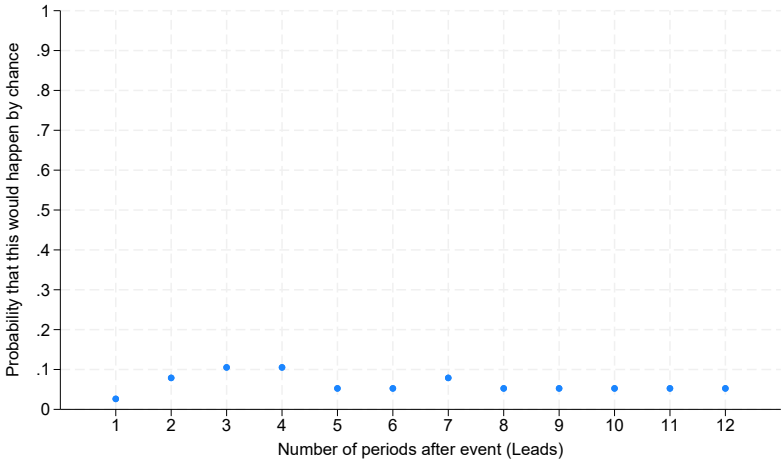

Figure 6: p-value plot

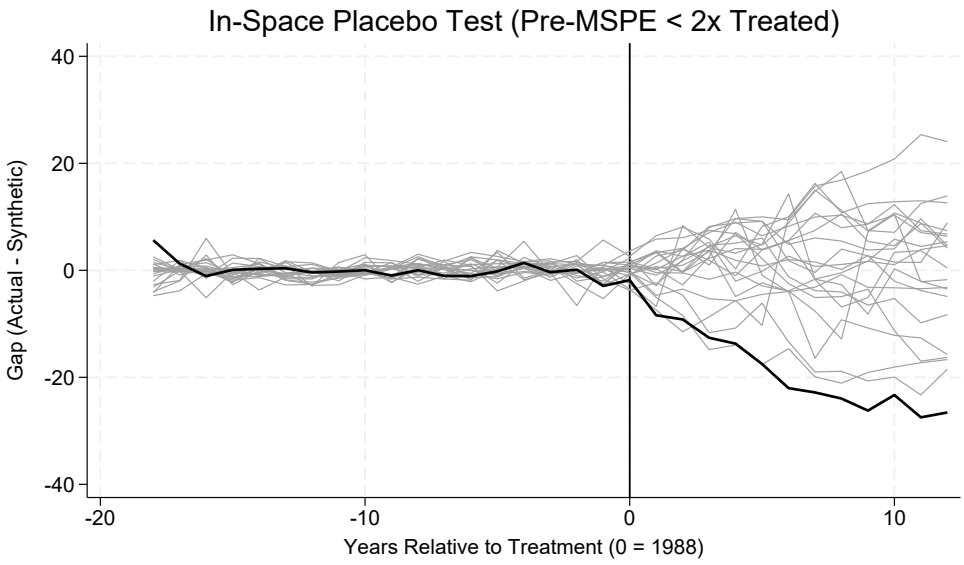

Figure 7: Placebo Plots (less than two times pre-treatment MSPE)

5.3 In-Time Placebo Model

Doing placebo testing with our fake treatment timing, we see from Figure 8 that the p-values are very large, signifying that the treatment was not significant. I also plot the placebos in Figure 9, which shows us that the real California is in the middle of the placebos. I did not restrict this plot to less than two times the real California's pre-treatment MSPE because if you do, you only have one placebo left. Both of these figures confirm this fake treatment has no significant effect on California's cigarette consumption.

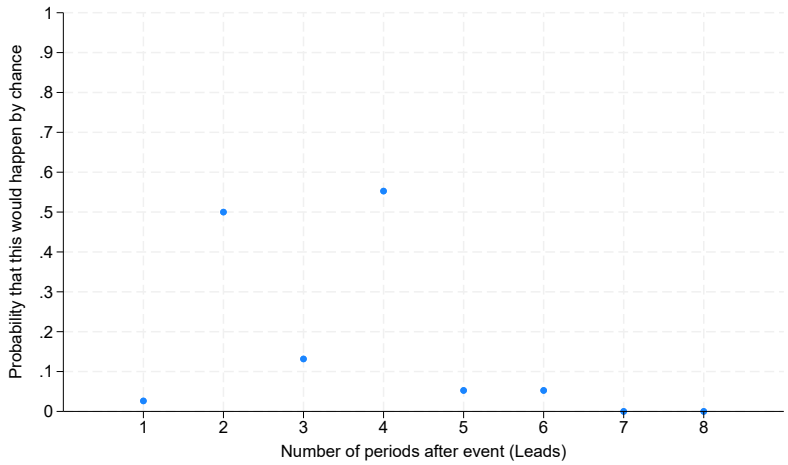

Figure 8: p-value plot

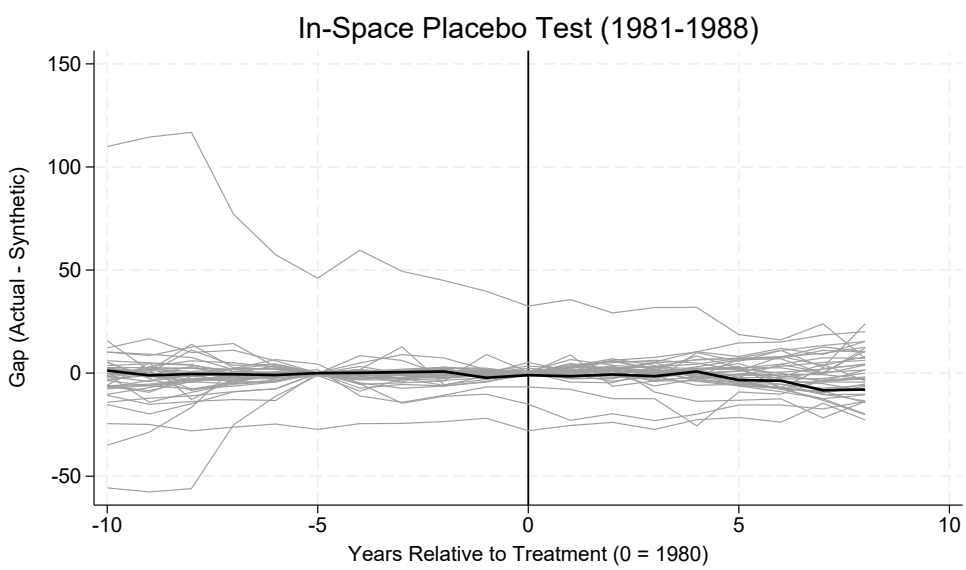

Figure 9: Placebo plots